



DIVAMICS

AI-Driven Drug Discovery with
Molecular Precision

DVI-001

IL-23R cyclopeptide inhibitor project

JNJ-2113 Pros and Cons

Major Advantages

High affinity in the low-picomolar range ($K_d=7.1\text{pM}$), selective blockade of IL-23R

- Oral cyclic peptides with gastrointestinal stability and systemic absorption
- $t_{1/2}$ has been extended to 9-16 hours, representing a significant improvement over first-generation therapeutic

在低皮摩尔范围内具有高亲和力 ($K_d=7.1\text{pM}$)，选择性阻断IL-23R

- 具有胃肠道稳定性和全身吸收的口服环肽
- $T_{1/2}$ 已延长至9-16小时，与第一代治疗相比有显著改善

Disadvantages

Clinical subjects are required to fast for ≥ 2 hours prior to dosing and for ≥ 0.5 hours post-dose (indicating significant room for PK optimization)

临床受试者在给药前禁食 ≥ 2 小时，给药后禁食 ≥ 0.5 小时（表明有很大的PK优化空间）

Our Goal

Implementing focused structural optimizations on the JNJ-2113 framework to develop analogs that preserve equivalent potency while significantly **boosting PK parameters**, with a primary focus on **increasing systemic AUC and oral bioavailability**.

对JNJ-2113框架进行重点结构优化，以开发在保持等效效力的同时显著**提高PK参数**的类似物，主要侧重于**提高系统AUC和口服生物利用度**。

Summary - 亲和力和体外活性与JNJ2113相当，静脉/口服的 PK上优于JNJ2113



Binding Affinity and In Vitro Activity

Compound	Target Binding vs. JNJ-2113	Binding Affinity and In Vitro Activity
DVC-P1-327	Comparable	DVC-P1-327 achieves picomolar-level target affinity and in vitro potency, consistent with the profile of JNJ-2113.

PK

Compound	IV PK vs. JNJ-2113	Oral PK vs. JNJ-2113
DVC-P1-327	Superior	Significant Superiority: Prolonged Oral Half-Life and Near-Doubling of AUC

DVC-P1-327 matches the high affinity and in vitro activity of JNJ-2113 but significantly outperforms it in pharmacokinetic properties via both IV and oral routes. Based on these findings, our preliminary project milestones have been fulfilled.

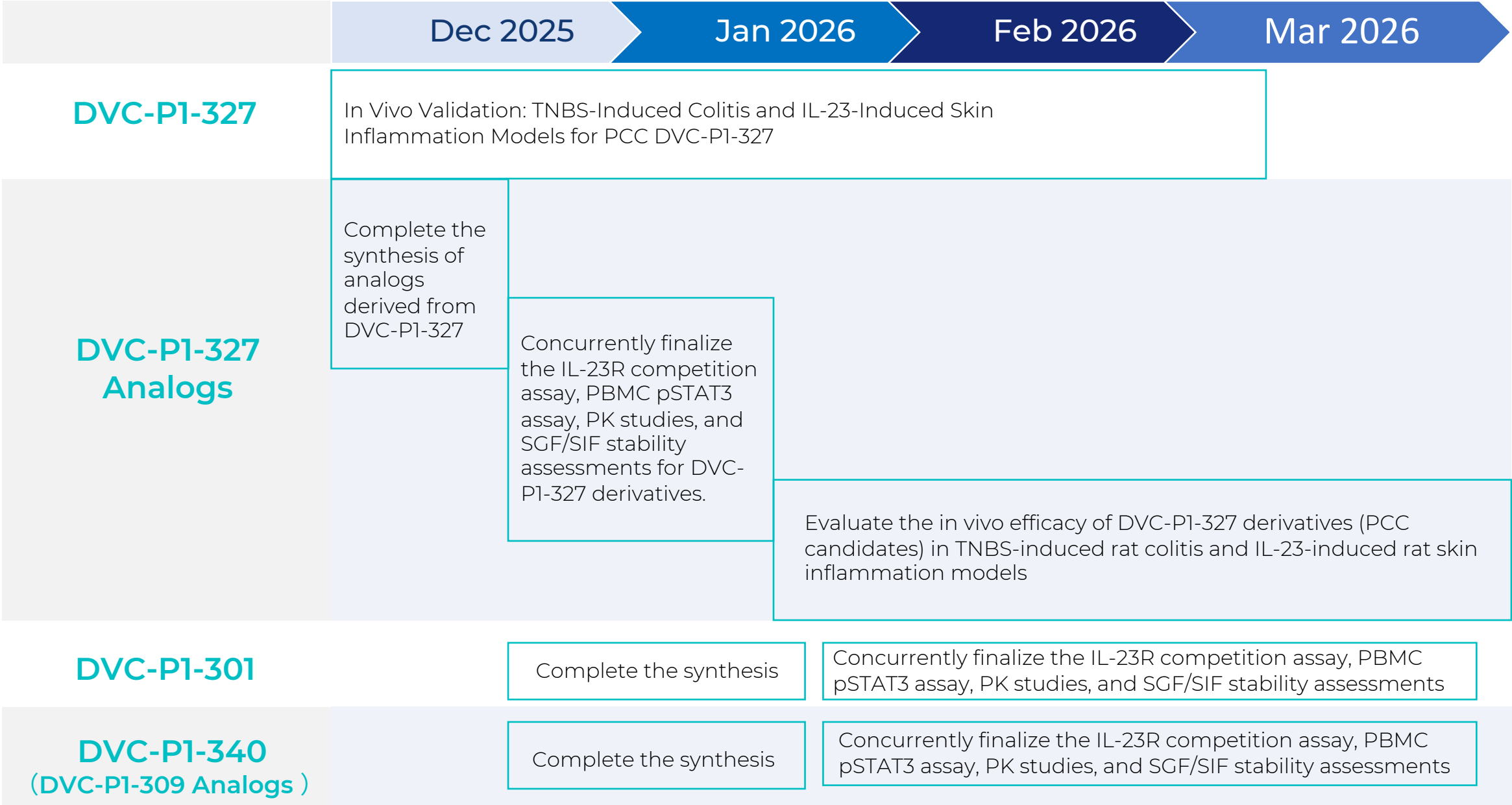
DVC-P1-327的高亲和力和体外活性与JNJ-2113相当，但在静脉和口服两种途径的药代动力学特性上都明显优于JNJ-2113。基于这些发现，我们初步的项目里程碑已经完成。

Target Product Profile for PCC of an Oral Peptide Targeting



Ideal target results (better than JNJ-2113)	Minimal target results (comparable)	JNJ-2113	DVC-P1-327
Binding affinity to hIL-23 (ELISA) KD: < 1 nM	KD: 1-10 nM	IC50=5.565 nM	IC50=4.809 nM
Bingding affinity to hIL-23R ECD (SPR) KD: < 1 pM (Cross species binding: rat IL-23R ECD binding KD: 1-10 pM)	KD: 1-10 pM (rat KD: 10-30 pM)	Human: KD=7.49 pM Rat: KD=1.49 pM	Human: KD=2.77 pM Rat: KD=1.20 pM
<i>In vitro</i> cellular based assays: - Human PBMC IL-23-induced pSTAT3 assay: IC50 1-5 pM - Human PBMC IL-12-induced pSTAT4 assay: IC50 > 5 µM (selectivity) - Human NK cells IL-23-induced IFNγ release: IC50 1-10 pM and/or - Human whole blood IL-23-induced IFNγ release: IC50 1-5 pM (Rat whole blood IL-23-induced IL-17A release)	<i>In vitro</i> cellular based assays: - IC50 5-50 pM - IC50 > 1 µM - IC50 5-50 pM - IC50 5-50 pM (Cross species activity: Rat whole blood assay)	- IC50= 10.8 pM - IC50 > 5 µM - IC50= 26.72 pM - IC50= 3.046 nM - IC50= 73 pM	- IC50=10.3 pM - IC50 > 5 µM - IC50= 164.54 pM - IC50 > 10 nM - IC50= 611 pM
ADME profiles: - T1/2 in FaSSIF: stable in 120 min - Liver or Kidney S9 matabolic stability assay: stable in 120 min - Unbounded% > 20% - Low DDI risk: No CYP inhibition @ 10 µM	ADME profiles: - T1/2 > 120 min - T1/2 > 120 min - Unbounded% 5-20% - CYP inhibition @ 10µM: < 50%	- stable in 120 min - stable in 120 min ≥20% ≤25%	- stable in 120 min - stable in 120 min ≥20% ≤10%
<i>In vivo</i> PK profiles: CL < 0.3 Qh across species; F% > 0.5%; moderate-long T1/2 Rat PK/PD profile: better than JNJ-2113 in ex vivo WB IL-23-stimulated IL-17A production	<i>In vivo</i> PK profiles: CL < 0.7 Qh across species; F% 0.1-0.3%; moderate T1/2 Rat PK/PD profile: comparable with JNJ-2113	<i>In vivo</i> PK profiles (mouse) Cl=6.44 mL/min/kg F%=0.626% T1/2=1.65 h <i>In vivo</i> PK profiles (Rat) Cl=6.20 mL/min/kg F%=0.0332% T1/2=3.28 h	<i>In vivo</i> PK profiles (mouse) Cl=10.6 mL/min/kg F%=0.528% T1/2=0.942 h <i>In vivo</i> PK profiles (Rat) Cl=6.61 mL/min/kg F%=0.0737% T1/2=3.29 h
<i>In vivo</i> efficacy: significant efficacy with clear dose response in rat IL-23-induced skin inflammation model and TNBS model, better than SoC and JNJ-2113	<i>In vivo</i> efficacy: comparable with SoC and JNJ-2113, or better efficacy on top of SoC		In rat IL-23-induced skin inflammation model 1 & 3 mg/kg: Superior efficacy observed compared to JNJ-2113. 10 mg/kg: Efficacy reaches a plateau, showing limited incremental response relative to JNJ-2113.
Non-GLP DRF in rats or monkey: Safety margin > 30 folds	Non-GLP DRF in rats or monkey: Safety margin > 10 folds		
Developability: kinetic solubility > 300µM, no risk in CMC and formulation	Developability: Controllable risk in CMC and formulation		
Solid FTO with desirable IP position	Solid FTO low IP risk		

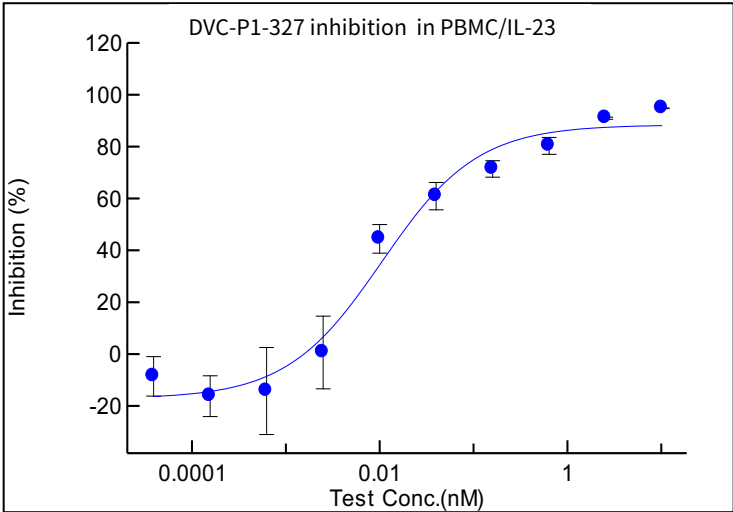
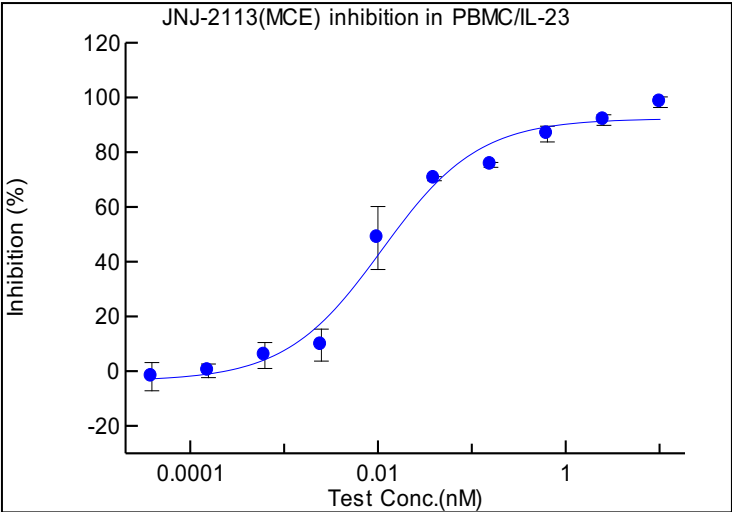
Experimental Timeline



Comparable in vitro activity in IL-23 mediated downstream signals



	K _D (SPR in vitro)		Human PBMC (IC ₅₀)		Human NK cells	Human whole blood	Rat whole blood
	Human IL-23R	Rat IL-23R	IL-23-induced pSTAT3 (pM)	IL-12-induced pSTAT4 (pM)	IL-23-induced IFN γ release (pM)	IL-23-induced IFN γ release (nM)	IL-23-induced IL-17A release (pM)
JNJ-2113(Ref)	7.49	1.49	10.8	>5,000,000	26.72	3.0406	73
DVC-P1-327	2.77	1.20	10.3	>5,000,000	164.54	>10	611



Chemical stability (FaSSIF)

Table 1. Stability results of test compounds and control compound in FaSSIF

Compound ID	Matrix	T _{1/2} (min)	Remaining Percentage (%)				
			0 min	15 min	30 min	60 min	120 min
DVC-P1-327	FaSSIF	> 511.69	100.00	106.71	100.90	104.88	105.15
JNJ-2113	FaSSIF	> 511.69	100.00	100.14	102.63	96.97	104.96

Note: If % remaining at 120 minutes was greater than 85%, then t_{1/2} will be reported as “> 511.69”

Similar to JNJ-2113, DVC-P1-327 is stable in FaSSIF within 120 min.

Low DDI risk



Table 2. Inhibition of 9 CYP Isoforms by Test Compounds in Human Liver Microsomes							
Compound ID	Concentration (µM)	% Inhibition					
		1A2	2C9	2C19	2D6	3A-M	3A-T
Alpha-Naphthoflavone	0.012	55.75	-	-	-	-	-
Sulfaphenazole	0.8	-	60.36	-	-	-	-
Loratadine	0.4	-	-	72.68	-	-	-
Quinidine	0.024	-	-	-	52.53	-	-
Ketoconazole	0.12	-	-	-	-	76.52	79.33
DVC-P1-327	10	9.57	-18.02	-10.77	-0.12	-3.09	4.08
JNJ-2113	10	8.57	25.40	3.72	0.54	3.87	-8.52

Compound DVC-P1-327 exhibits a safety profile comparable to JNJ-2113, with no apparent CYP inhibitory activity at 10 µM and a minimal risk of clinical drug-drug interactions

Liver S9 metabolic stability



Table 3. Metabolic Stability of Test Compounds in Pooled Male Rat and Human of Liver S9 Fractions (b)

Compound ID	Species	Assay Format	Percentage Remaining (%)				
			0 min	15 min	30 min	60 min	120 min
DVC-P1-327	Rat	With NADPH	100.00	90.58	101.57	99.15	99.28
DVC-P1-327	Rat	Without NADPH	100.00	95.05	94.82	96.17	97.07
DVC-P1-327	Human	With NADPH	100.00	94.13	94.51	92.64	107.29
DVC-P1-327	Human	Without NADPH	100.00	92.78	88.66	89.69	101.55
JNJ-2113	Rat	With NADPH	100.00	88.56	85.59	102.92	98.77
JNJ-2113	Rat	Without NADPH	100.00	86.37	98.64	100.68	106.81
JNJ-2113	Human	With NADPH	100.00	92.23	94.71	102.09	105.84
JNJ-2113	Human	Without NADPH	100.00	110.60	111.31	115.90	98.94

Similar to JNJ-2113, compound DVC-P1-327 exhibits no significant metabolism in liver S9 over a 120-minute incubation period.

Kidney S9 metabolic stability



Table 4. Metabolic Stability of Test Compounds in Pooled Male Rat and Human of Kidney S9 Fractions (b)

Compound ID	Species	Assay Format	Percentage Remaining (%)				
			0 min	15 min	30 min	60 min	120 min
DVC-P1-327	Rat	With NADPH	100.00	98.91	94.56	101.18	96.02
DVC-P1-327	Rat	Without NADPH	100.00	97.37	96.49	91.67	85.96
DVC-P1-327	Human	With NADPH	100.00	101.75	96.99	99.35	94.70
DVC-P1-327	Human	Without NADPH	100.00	100.58	109.30	104.94	101.74
JNJ-2113	Rat	With NADPH	100.00	101.87	99.24	102.32	100.79
JNJ-2113	Rat	Without NADPH	100.00	101.21	100.00	95.55	99.60
JNJ-2113	Human	With NADPH	100.00	98.36	91.56	99.17	98.00
JNJ-2113	Human	Without NADPH	100.00	100.38	96.24	92.86	101.13

Similar to JNJ-2113, compound DVC-P1-327 exhibits no significant metabolism in kidney S9 over a 120-minute incubation period.

Plasma proteins binding assay



Table 5. Protein Binding Results of Test Compound(s) in Rat and Human Plasma

Compound ID	Species	Test Concentration (μM)	% Unbound	% Bound	% Recovery	% Stability at 6 hrs
DVC-P1-327	Rat	1	21.69	78.31	111.97	105.03
DVC-P1-327	Human	1	24.94	75.06	88.57	102.02
JNJ-2113	Rat	1	26.53	73.47	115.81	115.33
JNJ-2113	Human	1	21.33	78.67	108.74	119.24

Similar to JNJ-2113, DVC-P1-327 unbound percentage is more than 20% in human and rat.

Pharmacokinetic Parameters via IV Dosing			
IV Dose: 1 mg/kg			
	Unit	Rat	
		JNJ-2113	DVC-P1-327
CL _{obs}	mL/min/kg	6.20	6.61
T _{1/2}	h	1.54	2.60
C ₀	ng/mL	3925	3971
AUC _{last}	h*ng/mL	2717	2512
AUC _{Inf}	h*ng/mL	2739	2531
AUC_%Extrap_obs	%	0.79	0.789
MRT _{Inf_obs}	h	1.07	1.16
AUC _{last} /D	h*mg/mL	2717	2512
V _{ss_obs}	L/kg	0.393	0.458

Following rational design iterations, DVC-P1-327 shows a **significantly extended T_{1/2}** while maintaining comparable IV pharmacokinetics and **similar** total systemic exposure (AUC) to JNJ-2113 in vivo.

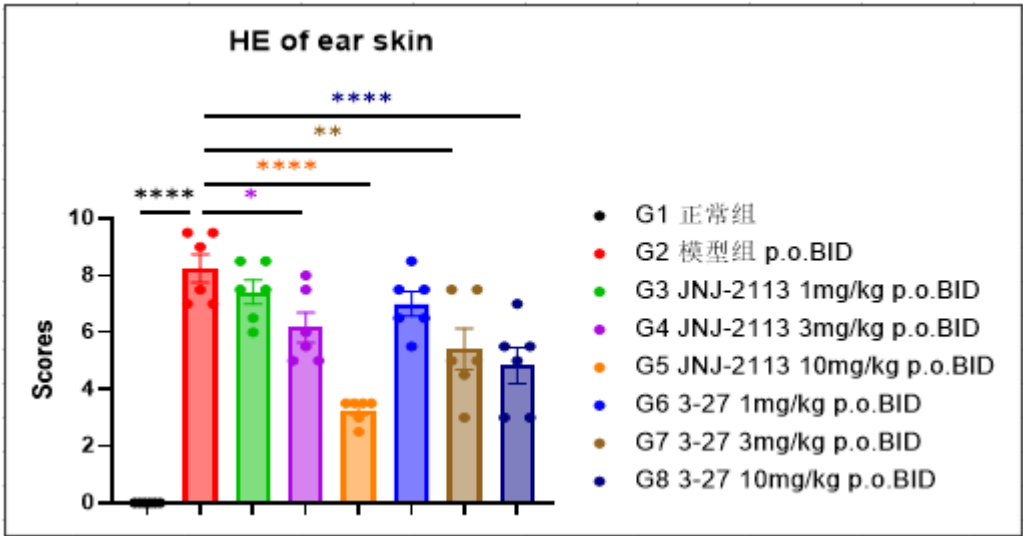
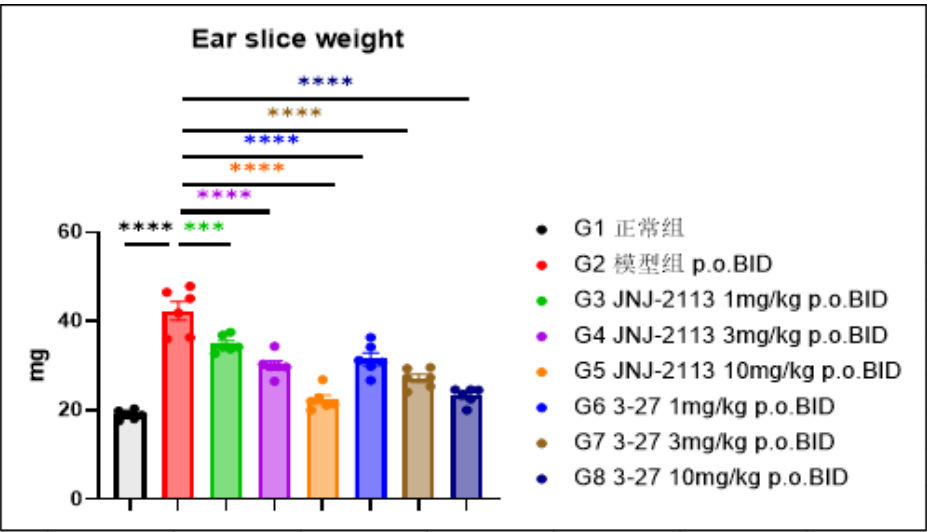
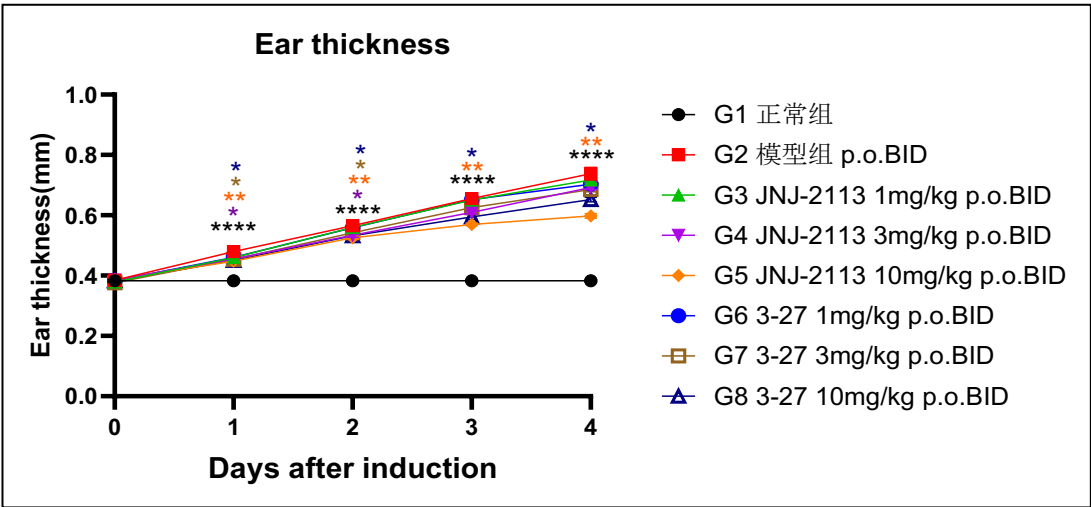
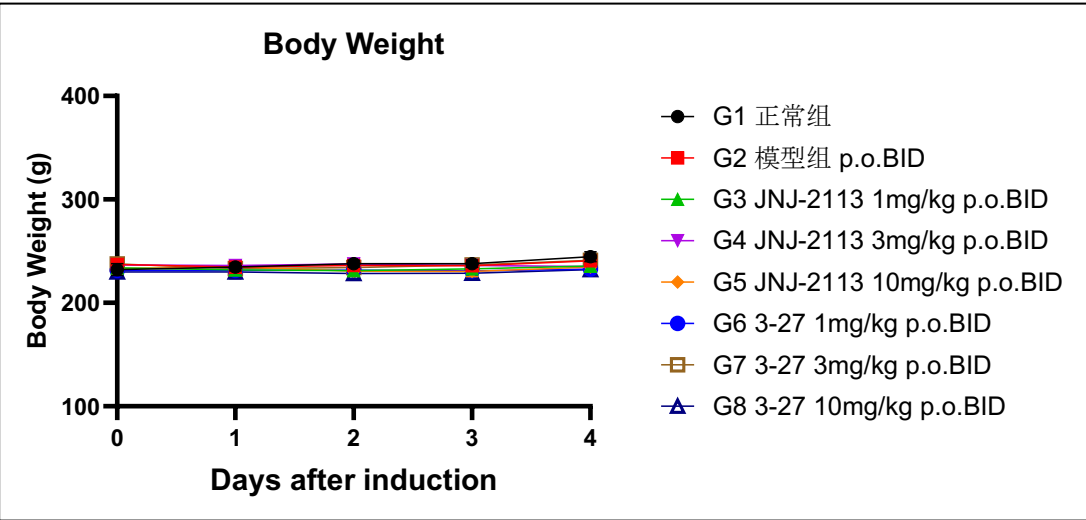
In Vivo PK



PO Dose: 10 mg/kg		Pharmacokinetic Parameters via PO Dosing	
		Rat	
	Unit	JNJ2113	DVC-P1-327
TI/2	h	3.28	3.29
Tmax	h	1.00	1.17
Cmax	ng/mL	3.96	5.28
AUClast	h*ng/mL	9.0	16.8
AUCInf	h*ng/mL	20.4	26.2
AUC_%Extrap_obs	%	53.6	35.9
MRTInf_obs	h	5.07	5.24
AUClast/D	h*mg/mL	0.90	1.68

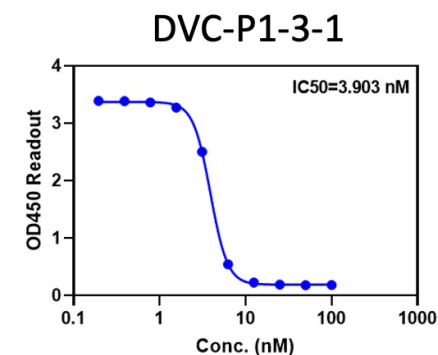
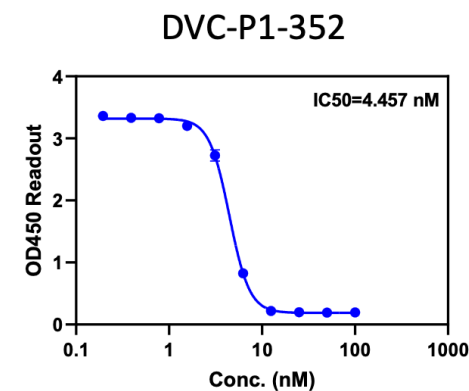
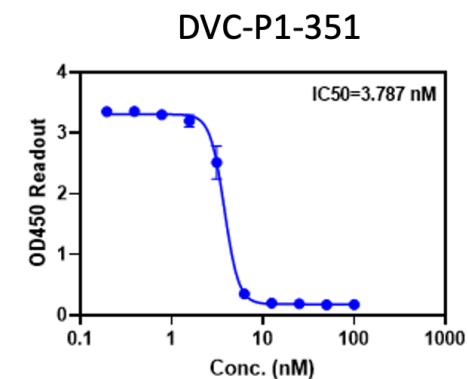
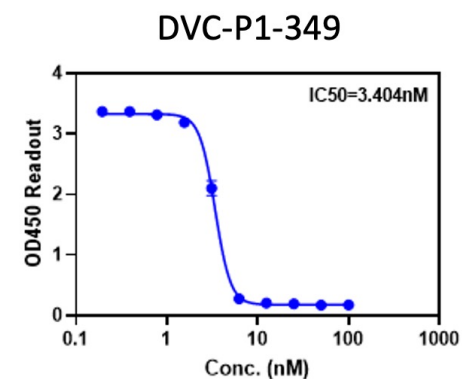
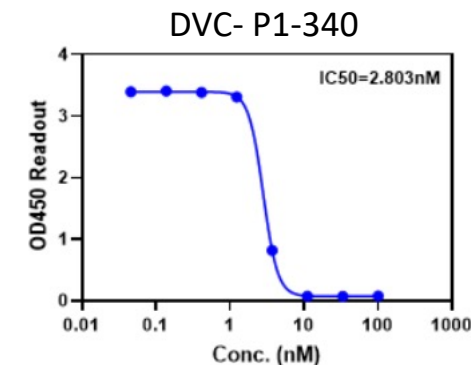
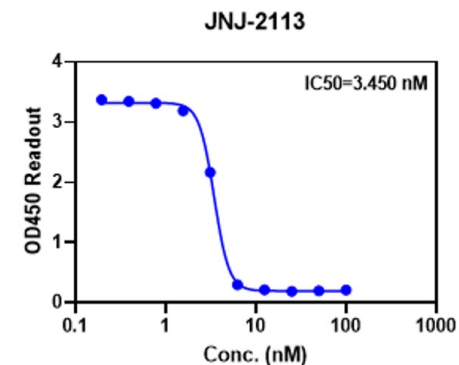
DVC-P1-327 demonstrates a comparable oral half-life to JNJ-2113, while achieving nearly 2-fold total drug exposure (AUC) in vivo.

Rat IL-23-induced psoriasis model

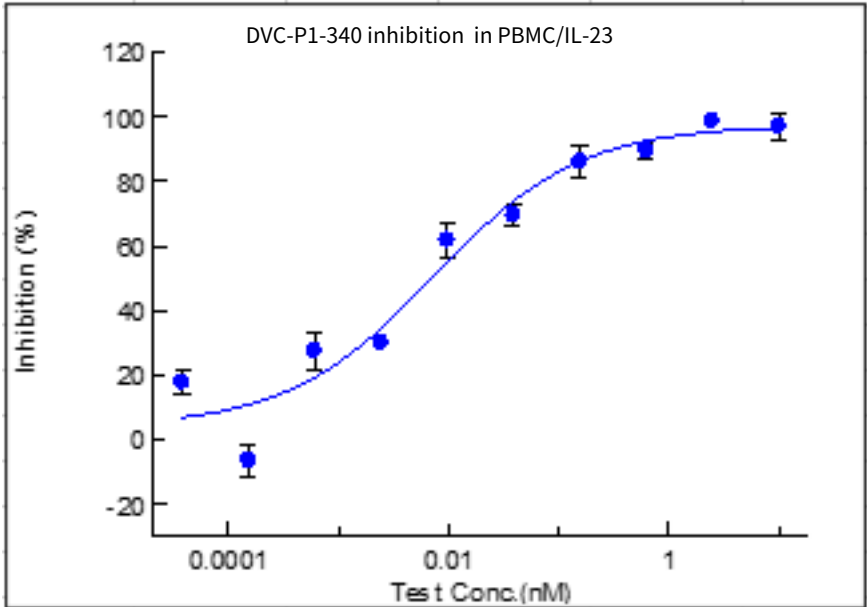
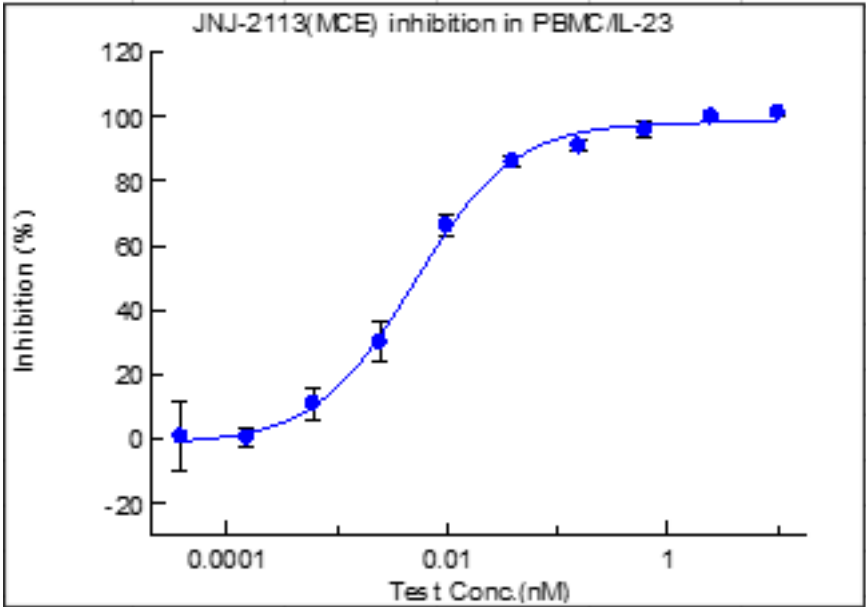


More candidates (more data ongoing testing)

Binding Affinity ELISA IC50, nM	
JNJ-2113(MCE)	3.450
DVC-P1-340	2.803
DVC-P1-349	3.404
DVC-P1-351	3.787
DVC-P1-352	4.457
DVC-P1-301	3.903



	K _D (SPR in vitro)		Human PBMC (IC ₅₀)	
	Human IL-23R	Rat IL-23R	IL-23-induced pSTAT3 (pM)	IL-12-induced pSTAT4 (pM)
JNJ-2113(Ref)	7.27	8.50	5.02	>5,000,000
DVC-P1-340	29.74	7.80	7.38	>5,000,000



DVC-P1-349/ DVC-P1-351 /DVC-P1-352/ DVC-P1-301



	K _D (SPR in vitro)	
	Human IL-23R	Rat IL-23R
JNJ-2113(Ref)	3.65	16.30
DVC-P1-349	5.74	37.10
DVC-P1-351	14.30	35.40
DVC-P1-352	14.50	30.00
DVC-P1-301	3.34	23.59

In Vivo PK (大鼠)

Summary of IV pharmacokinetic parameters

IV Dose 1 mg/kg		JNJ-2113		DVC-P1-327		DVC-P1-349		DVC-P1-351		DVC-P1-352		DVC-P1-340	
PK parameters	Unit	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Cl_obs	mL/min/kg	6.20	1.11	6.61	0.562	2.92	0.505	6.01	1.16	4.11	0.0593	7.08	0.344
T _{1/2}	h	1.54	0.09	2.60	2.10	1.07	0.205	0.838	0.0603	3.39	0.367	3.57	0.200
C ₀	ng/mL	3925	288	3971	532	7783	1733	4507	633	4683	626	2341	509
AUC _{last}	h*ng/mL	2717	444	2512	210	5803	1103	2833	500	4049	60.7	2353	116
AUC _{Inf}	h*ng/mL	2739	449	2531	205	5826	1103	2835	500	4056	59.0	2359	115
AUC_%Extrap_obs	%	0.79	0.19	0.787	0.356	0.397	0.150	0.0709	0.0115	0.188	0.0907	0.265	0.0536
MRT _{Inf_obs}	h	1.07	0.13	1.16	0.285	1.21	0.0414	0.812	0.0744	1.95	0.0537	2.21	0.0885
AUC _{last} /D	h*mg/mL	2717	444	2512	210	5803	1103	2833	500	4049	60.7	2353	116
V _{ss_obs}	L/kg	0.393	0.037	0.458	0.0894	0.211	0.0306	0.290	0.0319	0.480	0.0192	0.937	0.0636

In Vivo PK (大鼠)

Summary of PO pharmacokinetic parameters

PO Dose 10 mg/kg													
PK parameters	Unit	JNJ-2113		DVC-P1-327		DVC-P1-349		DVC-P1-351		DVC-P1-352		DVC-P1-340	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
T _{1/2}	h	3.28	0.57	3.29	1.30	2.15	NA	2.01	0.348	2.50	1.12	3.56	0.769
T _{max}	h	1.00	0.87	1.17	0.764	2.33	1.53	1.17	0.764	2.67	1.15	1.50	0.866
C _{max}	ng/mL	3.96	0.94	5.28	1.45	6.11	2.44	8.41	1.79	5.81	2.79	3.46	1.57
AUC _{last}	h*ng/mL	9.0	2.3	16.8	5.74	29.0	12.2	23.3	4.63	23.3	11.5	13.1	7.27
AUC _{Inf}	h*ng/mL	20.4	4.2	26.2	6.11	44.2	NA	27.1	4.46	30.3	11.5	21.4	10.0
AUC_%Extrap_obs	%	53.6	17.6	35.9	15.3	19.4	NA	14.3	3.30	25.1	10.1	40.3	6.95
MRT _{Inf_obs}	h	5.07	0.78	5.24	2.15	4.76	NA	3.19	0.265	4.76	0.998	5.79	1.22
AUC _{last} /D	h*mg/mL	0.90	0.23	1.68	0.574	2.90	1.22	2.33	0.463	2.33	1.15	1.31	0.727
F	%	0.0332	0.0085	0.0737	0.0336	0.0597	0.0296	0.0957	0.0157	0.0635	0.0384	0.0558	0.0309

In Vivo PK (小鼠)

Summary of IV pharmacokinetic parameters

IV Dose 1 mg/kg					
PK parameters	Unit	JNJ-2113		DVC-P1-3-1	
		Mean	SD	Mean	SD
Cl_obs	mL/min/kg	6.44	0.726	0.335	0.0303
T _{1/2}	h	1.21	0.0608	3.62	0.204
C ₀	ng/mL	4360	565	8960	1093
AUC _{last}	h*ng/mL	2603	307	49631	4296
AUC _{Inf}	h*ng/mL	2612	307	50057	4330
AUC_%Extrap_obs	%	0.339	0.0454	0.851	0.236
MRT _{Inf_obs}	h	0.866	0.0504	5.01	0.169
AUC _{last} /D	h*mg/mL	2603	307	49631	4296
V _{ss_obs}	L/kg	0.333	0.0190	0.101	0.00790

Summary of PO pharmacokinetic parameters

PO Dose 10 mg/kg					
PK parameters	Unit	JNJ-2113		DVC-P1-3-1	
		Mean	SD	Mean	SD
T _{1/2}	h	1.65	0.0961	4.49	0.106
T _{max}	h	1.00	0	2.67	1.15
C _{max}	ng/mL	53.2	22.5	205	118
AUC _{last}	h*ng/mL	139	47.0	1858	1780
AUC _{Inf}	h*ng/mL	164	66.4	2202	1598
AUC_%Extrap_obs	%	13.3	5.51	24.4	18.6
MRT _{Inf_obs}	h	2.77	0.174	7.40	0.360
AUC _{last} /D	h*mg/mL	13.9	4.70	186	178
F	%	0.626	0.254	0.380	0.369

注：因目前DVC-P1-3-1化合物合成量不足以支撑大鼠PK，故先进行小鼠PK实验

Suzhou (Headquarters) | Shanghai | Beijing | Singapore

THANK YOU !

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